

Live Market Regime Identifier for MetaTrader 5 (MT5)

This Python GUI application connects to your MetaTrader 5 terminal, fetches live XAUUSD H1 data, applies a pre-trained machine learning model to identify the current market regime, and displays it in real-time.

Features

- Connects to MetaTrader 5 for live data.
- Fetches XAUUSD H1 OHLCV data.
- Applies feature engineering (returns, volatility, moving averages, RSI, ATR).
- Uses a pre-trained Gaussian Mixture Model (GMM) to predict market regimes.
- Displays the current market regime and price in a user-friendly GUI.
- Automatically refreshes data and predictions at a configurable interval.

Prerequisites

Before running the application, ensure you have the following:

1. **MetaTrader 5 (MT5) Terminal:** Installed and running on your system. Make sure you are logged into your trading account.
2. **Python 3.x:** Installed on your system.
3. **Trained Models:** The `market_regime_gmm.pkl` (trained GMM model) and `scaler.pkl` (StandardScaler) files must be in the same directory as the `mt5_regime_gui.py` script. These files are generated by the previous market regime identification process.

Installation

1. **Download the files:** Ensure `mt5_regime_gui.py`, `market_regime_gmm.pkl`, and `scaler.pkl` are in the same directory.
2. **Install Python Libraries:** Open your terminal or command prompt and run the following command to install the necessary Python packages:

```
Bash
```

```
pip install MetaTrader5 pandas numpy scikit-learn tkinter
```

Note: `tkinter` is usually included with Python installations. If you encounter issues, you might need to install it separately for your OS (e.g., `sudo apt-get install python3-tk` on Debian/Ubuntu).

MT5 Configuration

To allow Python to connect to MT5, you need to enable "Allow DDE server" and "Allow automated trading" in your MT5 terminal:

1. Open your MT5 terminal.
2. Go to `Tools` -> `Options` .
3. Navigate to the `Server` tab and ensure `Allow DDE server` is checked.
4. Navigate to the `Expert Advisors` tab and ensure `Allow automated trading` is checked.
5. Restart your MT5 terminal if you made any changes.

Usage

1. **Ensure MT5 is running:** Make sure your MetaTrader 5 terminal is open and logged into an account.
2. **Run the Python script:** Open your terminal or command prompt, navigate to the directory where you saved the files, and run:

```
Bash
```

```
python mt5_regime_gui.py
```

3. A GUI window will appear, displaying the current XAUUSD price and the identified market regime. The data will refresh automatically every 60 seconds.

Customization

You can modify the following parameters at the beginning of the `mt5_regime_gui.py` script:

- `SYMBOL` : The trading symbol (e.g., "XAUUSD").
- `TIMEFRAME` : The MT5 timeframe (e.g., `mt5.TIMEFRAME_H1` for 1-hour).
- `DATA_BARS` : Number of historical bars to fetch for feature calculation.
- `REFRESH_INTERVAL_SECONDS` : How often the data and regime prediction should refresh.

Troubleshooting

- **"MT5 Connection Error"**: Ensure MT5 is running, you are logged in, and the MT5 configuration steps (Allow DDE server, Allow automated trading) have been followed.
- **"Trained models not found"**: Make sure `market_regime_gmm.pkl` and `scaler.pkl` are in the same directory as the script.
- **"Not enough data for features"**: Increase `DATA_BARS` or ensure enough historical data is available in your MT5 terminal for the selected symbol and timeframe.

Disclaimer

This tool is for educational and informational purposes only and should not be considered financial advice. Trading in financial markets carries a high level of risk, and you may lose money. Always consult with a qualified financial professional before making any trading decisions.